

The quadratic quantum Wasserstein channel complexity is not convex

Candidate counterexample for arXiv:2402.13150
Run `fa_banach_001`, lane 10, model GPT5.6

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Status. Candidate counterexample likely valid; full negative answer to the channel-complexity convexity question in Section 5.1 of arXiv:2402.13150. On one qubit, for the source paper’s symmetric Pauli cost, the mixture of the identity channel and the Pauli- X channel violates the convexity inequality. All estimates are exact and finite-dimensional.

1 Source question

Bunth, Pitrik, Titkos, and Virosztek define, for a finite collection of observables $\mathcal{A} = \{A_1, \dots, A_k\}$, the channel Wasserstein complexity

$$C_W(\Phi) = \max_{\rho \in \mathcal{S}(\mathcal{H})} d_{\mathcal{A}}(\rho, \Phi(\rho)), \quad (1)$$

where $d_{\mathcal{A}}$ is their modified quadratic quantum Wasserstein divergence [1]. At the end of Section 5.1 (printed page 18), they write:

“We note that whether the Wasserstein complexity (37) is convex or not is an open question.”

The original passage is reproduced in Figure 1.

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where the first inequality follows from the triangle inequality for the Wasserstein divergence $d_{\mathcal{A}}$, the second inequality is due to maximizing terms of a sum separately, and the last inequality follows from broadening the domain of the second maximum. From this it also follows directly that the Wasserstein complexity is subadditive under tensor products in the following sense:

$$C_W(\Phi_1 \otimes \Phi_2) \leq C_W(\Phi_1 \otimes I) + C_W(I \otimes \Phi_2).$$

We note that whether the Wasserstein complexity (37) is convex or not is an open question.

5.2. Mean-field approximations of evolution equations. Among the many applications of classical optimal transport, one is particularly important for the development of quantum optimal transport theory. In [20], Dobrushin employed a special approximation based on a transport-related matrix, the so-called Kantorovich–Rubinstein matrix, to prove the uniqueness of the solution to the

Figure 1: The open question in arXiv:2402.13150v4, printed page 18.

Here convexity has its standard meaning on the convex set of quantum channels:

$$C_W((1-t)\Phi_0 + t\Phi_1) \leq (1-t)C_W(\Phi_0) + tC_W(\Phi_1), \quad 0 \leq t \leq 1.$$

2 Definitions and proof intuition

For two states ρ, ω , let

$$D_{\mathcal{A}}^2(\rho, \omega) = \min_{\Pi \in \mathcal{C}(\rho, \omega)} \text{Tr}(C_{\mathcal{A}} \Pi), \quad C_{\mathcal{A}} = \sum_j (A_j \otimes I - I \otimes A_j^T)^2,$$

with the coupling convention of [1]. The modified divergence is

$$d_{\mathcal{A}}^2(\rho, \omega) = D_{\mathcal{A}}^2(\rho, \omega) - \frac{1}{2}(D_{\mathcal{A}}^2(\rho, \rho) + D_{\mathcal{A}}^2(\omega, \omega)). \quad (2)$$

Proof intuition. Take the source paper's own symmetric qubit cost $\mathcal{A}_s = \{\sigma_x, \sigma_y, \sigma_z\}$. Mix the identity channel with the bit flip $X(\rho) = \sigma_x \rho \sigma_x$. Convexity would make the complexity of this mixture grow at most linearly in the mixing parameter t . But on the pure input $|0\rangle\langle 0|$, the output immediately acquires a second eigenvalue t . The square root in the definition of d_s produces an exact term $\sqrt{t(1-t)}$ in d_s^2 , and hence $d_s \asymp t^{1/4}$. This is much larger than a linear function near zero.

Only one global estimate is needed: the Pauli cost operator equals $8(I - |\Omega\rangle\langle\Omega|)$, so every channel has complexity at most $2\sqrt{2}$. At $t = 1/4$, this upper bound on the convexity right-hand side is $\sqrt{2}/2$, while the single pure input gives the exact lower bound $\sqrt{1 + \sqrt{3}}$.

3 Two elementary Pauli calculations

Write

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and abbreviate $D_s = D_{\mathcal{A}_s}$ and $d_s = d_{\mathcal{A}_s}$ as in the source paper.

Lemma 1 (Pure-input formula). *If at least one of $\rho, \omega \in \mathcal{S}(\mathbb{C}^2)$ is pure, then*

$$d_s^2(\rho, \omega) = \sum_{j=x,y,z} \{ \text{Tr}(\sqrt{\rho} \sigma_j \sqrt{\rho} \sigma_j) + \text{Tr}(\sqrt{\omega} \sigma_j \sqrt{\omega} \sigma_j) - 2 \text{Tr}(\rho \sigma_j) \text{Tr}(\omega \sigma_j) \}. \quad (3)$$

Proof. When one marginal is pure, its only coupling with the other state is the product coupling. Expanding its cost gives

$$D_s^2(\rho, \omega) = \sum_j (\text{Tr}(\rho \sigma_j^2) + \text{Tr}(\omega \sigma_j^2) - 2 \text{Tr}(\rho \sigma_j) \text{Tr}(\omega \sigma_j)).$$

For any state τ , the canonical purification realizes the self-cost and gives

$$D_s^2(\tau, \tau) = 2 \sum_j (\text{Tr}(\tau \sigma_j^2) - \text{Tr}(\sqrt{\tau} \sigma_j \sqrt{\tau} \sigma_j)).$$

Substitution in (2) proves (3). These are also the source paper's displayed pure-state and self-cost formulas [1, Sec. 2]. $\square$

Lemma 2 (Uniform diameter bound). *For every pair of qubit states, $d_s(\rho, \omega) \leq 2\sqrt{2}$. Consequently, $C_W(\Phi) \leq 2\sqrt{2}$ for every qubit channel Φ .*

Proof. Let

$$|\Omega\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

The Pauli expansion of the maximally entangled projection is

$$\sum_{j=x,y,z} \sigma_j \otimes \sigma_j^T = 4|\Omega\rangle\langle\Omega| - I.$$

Since $\sigma_j^2 = I$,

$$\begin{aligned} C_s &= \sum_j (\sigma_j \otimes I - I \otimes \sigma_j^T)^2 \\ &= 6I - 2 \sum_j \sigma_j \otimes \sigma_j^T = 8(I - |\Omega\rangle\langle\Omega|). \end{aligned}$$

Thus $0 \leq C_s \leq 8I$. Every coupling has cost at most 8, so $D_s^2(\rho, \omega) \leq 8$. Formula (2) subtracts nonnegative self-costs; hence $d_s^2 \leq D_s^2 \leq 8$. Maximizing over ρ proves the channel bound. The same spectral decomposition appears in [1, Eq. (24)]. $\square$

4 The counterexample

Let X be the Pauli bit-flip channel and define

$$X(\rho) = \sigma_x \rho \sigma_x, \quad \Lambda_t = (1-t)\text{Id} + tX. \quad (4)$$

Theorem 1. *For the symmetric Pauli quadratic Wasserstein divergence on one qubit, the channel complexity (1) is not convex. Explicitly,*

$$C_W\left(\frac{3}{4}\text{Id} + \frac{1}{4}X\right) \geq \sqrt{1+\sqrt{3}} > \frac{\sqrt{2}}{2} \geq \frac{3}{4}C_W(\text{Id}) + \frac{1}{4}C_W(X).$$

Proof. Use the pure input

$$\rho_0 = |0\rangle\langle 0|, \quad \omega_t = \Lambda_t(\rho_0) = \begin{pmatrix} 1-t & 0 \\ 0 & t \end{pmatrix}.$$

Apply Lemma 1. For σ_z , the three terms in braces are

$$1 + 1 - 2(1)(1-2t) = 4t.$$

For each of σ_x and σ_y , both expectation values and the ρ_0 square-root term vanish, while

$$\text{Tr}(\sqrt{\omega_t} \sigma_j \sqrt{\omega_t} \sigma_j) = 2\sqrt{t(1-t)}.$$

Therefore

$$d_s^2(\rho_0, \Lambda_t(\rho_0)) = 4t + 4\sqrt{t(1-t)}, \quad C_W(\Lambda_t) \geq \sqrt{4t + 4\sqrt{t(1-t)}}. \quad (5)$$

At $t = 1/4$, this lower bound is $\sqrt{1+\sqrt{3}}$.

On the other hand, $C_W(\text{Id}) = 0$ because $d_s(\rho, \rho) = 0$. By Lemma 2, $C_W(X) \leq 2\sqrt{2}$. If C_W were convex, then

$$C_W(\Lambda_{1/4}) \leq \frac{3}{4}C_W(\text{Id}) + \frac{1}{4}C_W(X) \leq \frac{\sqrt{2}}{2}.$$

This contradicts (5), since $1 + \sqrt{3} > 1/2$. Hence C_W is not convex. $\square$

Remark 1 (Robust mechanism). Equation (5) gives $C_W(\Lambda_t) \geq 2t^{1/4}(1 + o(1))$ as $t \downarrow 0$, whereas convexity would give $C_W(\Lambda_t) \leq 2\sqrt{2}t$. Thus the violation holds for an interval of small positive t and is not a numerical coincidence at $t = 1/4$.

5 Verification, scope, and novelty

The proof is exact; no numerical optimization over couplings or states is used. The accompanying script `code/verify_counterexample.py` symbolically checks the Pauli cost identity, its spectrum, the three terms in (3), and the strict convexity gap at $t = 1/4$. It is a redundant finite-dimensional check, not a substitute for the proof.

The counterexample answers exactly the question as stated: the observable collection is the source paper’s standard symmetric Pauli choice, both endpoints and their mixture are quantum channels, and the same divergence and maximization as equation (37) are used. It does not assert that every other quantum Wasserstein channel complexity is nonconvex, nor does it address convexity notions for the different order-one/Hamming-Wasserstein complexity.

A bounded literature search was performed on 22 July 2026 using the exact source sentence, the source title and arXiv id, and combinations of “quantum Wasserstein complexity,” “convex,” “channel,” “Pauli,” and “quadratic divergence.” The local run indexes were also searched. No prior answer to this exact Section 5.1 question was found. Search hits about convexity concerned different order-one or resource-dependent channel complexities. Novelty is therefore plausible but not certified.

The source paper’s separate De Palma–Trevisan triangle-inequality conjecture is no longer open: Wirth subsequently proved it in full generality [2]. That literature update is why the present packet targets the still-distinct channel-complexity question rather than claiming a new proof of the paper’s main conjecture.

Human-review recommendation. High priority and low expected effort. Check (i) the pure-input formula’s normalization against equation (37), (ii) the identity $C_s = 8(I - |\Omega\rangle\langle\Omega|)$, and (iii) that convexity is meant on the affine space of channels. These three checks reproduce the entire argument.

References

- [1] G. Bunth, J. Pitrik, T. Titkos, and D. Virostek, *On the metric property of quantum Wasserstein divergences*, arXiv:2402.13150v4 (2025), especially Sections 2, 3.1, and 5.1.
- [2] M. Wirth, *Triangle Inequality for a Quantum Wasserstein Divergence*, arXiv:2511.20450 (2025).